

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2023/2024

MAY EXAMINATION

UCCD3053 INFORMATION TECHNOLOGY PROFESSIONAL ETHICS

THURSDAY, 23 MAY 2024

TIME : 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
BUSINESS INFORMATION SYSTEMS
BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
DIGITAL ECONOMY TECHNOLOGY

Instructions to Candidates:

This question paper consists of **THREE (3)** questions in **Section A** and **TWO (2)** questions in **Section B**.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**.
Each question carries 25 marks.

Should a candidate answer more than **ONE (1)** question in **Section B**, marks will only be awarded for the **FIRST** question in that section in the order the candidate submits the answers.

Candidates are not allowed to use calculator.

Answer questions only in the answer booklet provided.

UCCD3053 INFORMATION TECHNOLOGY PROFESSIONAL ETHICS

Section A: Answer ALL questions in this section.

- Q1. (a) Section 233 of the Communications and Multimedia Act 1998 (CMA) criminalizes online content that is obscene, indecent, false, offensive with intent to annoy, abuse, threaten or harass another person. State the punishment for any conviction under Section 233 of the CMA. (6 marks)
- (b) Section 233 of CMA prohibits any annoying, abusive, false or offensive posting of news on the **THREE (3) R's**. Explain your understanding of such offences. (9 marks)
- (c) The Personal Data Protection Act 2010 (PDPA) regulates the processing of all personal data in regards to commercial transactions. However, there are certain data such as cookies, geolocations and data mining that relates to online privacy which makes its enforcement challenging. Provide **FIVE (5)** areas of protection under the PDPA. (5 marks)
- (d) Discuss the penalty for non-compliance of the PDPA. (5 marks)
- [Total : 25 marks]
- Q2. (a) Professional Ethics in Information Technology (IT) ensures the proper behavior of an IT professional. Explain **ANY THREE (3)** ways each on how an IT Professional should act towards his:
- (i) Supplier (3 marks)
- (ii) Employer (3 marks)
- (iii) Clients (3 marks)
- (iv) Colleagues (3 marks)
- (b) You are the Head of Human Resources in an Information Technology (IT) company. Explain **THREE (3)** conflict of interest issues of each of the staff below that will make them liable for dismissal:
- (i) Software engineer (3 marks)
- (ii) Purchasing officer (3 marks)
- (iii) Web Developer (3 marks)

UCCD3053 INFORMATION TECHNOLOGY PROFESSIONAL ETHICS**Q2. (Continued)**

- (c) Explain **TWO (2)** aspects of Syariah law which is different from Secular law.
(4 marks)
[Total : 25 marks]

- Q3. (a) This news was taken from “The Malay Mail”, Tuesday 3rd January 2023. Read the article and answer the following questions:

KUCHING, January 3rd 2023 —

A housewife who responded to an advertisement posted on Facebook, lost RM44,074 to a syndicate through illegal transfers late last month, Sarawak DAP chairman Chong Chieng Jen said today. It all started when she surfed Facebook and came across an advertisement for a house cleaning service. As the festive period was approaching, she engaged the administrator of the Facebook page to book the house-cleaning service and upload all relevant information needed onto an APK link provided. BUT the housewife encountered problems when trying to make payment. The following day she discovered that her bank account was blocked and upon enquiry, she found that there had been eight consecutive transfers of funds to a 3rd party account with a different bank within an hour. What was peculiar was that she had set her account with a one-day maximum withdrawal limit of RM10,000, but she did not receive any notification from the bank that her account limit was altered. The victim did not receive any alert from the bank’s security or any One Time Password (OTP) from the bank.

- (i) Explain what is online fraud and the modus operandi used here.
(4 marks)
- (ii) State **TWO (2)** legal and ethical aspects of this case. (4 marks)
- (iii) Explain the actions to be taken and the roles and responsibilities of both:
- (a) Bank Negara (4 marks)
- (b) Malaysian Communications and Multimedia Commission.
(MCMC) (4 marks)
- (iv) Explain why such cases has been increasing recently despite repeated warnings from the authorities. (4 marks)
- (v) State the Law and penalties if a suspect is arrested, charged and found guilty. (5 marks)
[Total : 25 marks]

UCCD3053 INFORMATION TECHNOLOGY PROFESSIONAL ETHICS**Section B (ANSWER ONE (1) QUESTION)**

- Q4. (a) Copyright, Trademarks and Trade secrets are very important to Law and ethics in Information Technology. Discuss the consequences if an employee or another competitor infringes it. (6 marks)
- (b) GEELY is an automobile company, based in Hangzhou, China. Before it bought a major stake in PROTON Malaysia in June 2017, GEELY had also bought into the VOLVO automobile company of Sweden in 2010.
- (i) Explain the application of reverse engineering with Volvo XC90 sports utility vehicle (SUV) and the Proton X70 sports utility vehicle (SUV). Discuss its similar design in both SUV's from **THREE (3)** legal and ethical aspects. (6 marks)
- (ii) Explain **THREE (3)** differences outlined in the ZEUS Coffee vs ZUS Coffee copyright case. (6 marks)
- (c) Fair-Use is an exception in reproducing copyrighted works without consent.
- (i) State **FOUR (4)** situations that are considered as Fair-Use usage. (4 marks)
- (ii) Advise how a businessman can register and protect his trademark. (3 marks)
- [Total : 25 marks]

UCCD3053 INFORMATION TECHNOLOGY PROFESSIONAL ETHICS

- Q5. (a) Artificial Intelligence (AI) is the intelligence of machines or software, which is the opposite of human intelligence. Anyone looking to use machine learning as part of real world systems, needs to factor ethics into their AI training process and strive to avoid bias. Machine learning requires large amount of data. Techniques used to get such data have raised concerns on privacy, surveillance and copyright. Answer the following questions:
- (i) To build speech recognition algorithms, AMAZON has recorded millions of private conversations for workers to listen and transcribe for training purposes. Explain **THREE (3)** potential ethical breaches for such recordings. (6 marks)
 - (ii) In 2022, generative AI began to create images, audio, video and text that are similar to real photographs, recordings, films or human writing. Provide **FOUR (4)** abuses that have taken place. (8 marks)
 - (iii) Facial recognition, voice recognition and deep fake have cause several potential ethical abuse and misuse. Identify **THREE (3)** moral questions related to such AI misuse. (6 marks)
- (b) The Covid-19 Pandemic allowed many jobs to be done remotely. Working from home (Telework) was a norm then. List **FIVE (5)** advantages of Telework. (5 marks)
- [Total : 25 marks]
-

